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Determinants of Finger Choice in Piano Sight-Reading

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Sixteen pianists sight-read the unfingered right-hand score of 7 studies by Czerny. The pianists were of 3 levels of expertise. Each study was performed twice. Fingerings were transcribed from video recordings. Measures were taken of performance accuracy and fingering consistency. The choices made were compared to the predictions of a model based primarily on motor–anatomical considerations. Performance accuracy and fingering consistency were both correlated positively with expertise and negatively with the calculated difficulty of fingerings according to the model. These data are consistent with the notion that expertise in piano fingering depends on the availability of overlearned, rule-governed response sequences triggered by familiar visual patterns within the notation.

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The piano presents a unique motor programming problem when compared with almost any other motor task. The problem consists of deciding, often in extremely time-limited circumstances, which combinations of fingers to use (from a typically large set of possibilities) in executing a given sequence of notes.

In many other situations (e.g., wind instruments, typing), there is generally a one-to-one correspondence between a key and the finger normally used to depress that key. It may take only a few days for a motivated learner to master all the basic fingerings. Consequently, motor skill acquisition primarily involves perfecting faster and smoother transitions between these basic fixed positions.

In the case of the piano, a sounded note is defined by a particular key to be struck, but any of the 10 available fingers may be considered as potential candidates for executing the note. Therefore, there is no such thing as a standard fingering for a note. Everything depends on the context in which that note is found. As a consequence, skill in finger choice is something that develops over long periods of time and can be considered an example of cognitive expertise in problem solving. Given the obvious cognitive infeasibility of examining all alternative solutions algorithmically, fingering expertise must depend on the acquisition of heuristic devices for reducing the search space.

The bulk of the existing research literature on finger movements examines tasks in which finger choice is highly constrained or nonexistent (Rosenbaum, 1987; Semjen & Garcia-Colera, 1986), and this research therefore has limited relevance to piano playing. However, a number of authors (e.g., Bernstein, 1967; Kugler, Kelso, & Turvey, 1980; Rosenbaum, Vaughan, Jorgensen, Barnes, & Stewart, 1993) have shown how interactions or dependencies within the motor system can lead to there being fewer effective degrees of freedom than would be predicted from a simple count of the body's mechanical degrees of freedom.

Our approach to the problem of mapping the heuristic repertoire of pianists is to consider separately three different categories of potential constraints on the search space: motor–anatomical, cognitive, and artistic. Motor–anatomical constraints reflect anatomical properties of the hand (e.g., maximum possible and comfortable stretch between fingers), properties of the keyboard (e.g., black keys set above and behind white keys), and the motor independence and preferred direction of movement of the fingers. Cognitive constraints reflect properties of executive and memory processes. Within the broad category of cognitive constraints are some that have their effect primarily in the short term. For instance, fingering solutions that require the immediate repetition of previously used sequences may be preferred to those that require some change (cf. Rosenbaum, 1987). Or a fingering solution that entails preservation of finger–key mapping may be preferred to one that entails a recalibration. Other cognitive constraints may operate over longer time periods. For instance, it may aid long-term memorization if equivalent passages in different parts of an extended composition are fingered in the same way. Certain frequently occurring patterns (e.g., scales and broken chords) may have codified fingerings prescribed in pedagogic texts ("scale-book" fingerings) that are learned as default options and held in long-term memory for general use across a wide range of compositions. Artistic constraints reflect expressive requirements of the music such as maintaining an unbroken legato line or accenting a particular note.

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Fingering strategies have always been of intense interest to keyboard players because it is believed that fingering can significantly affect the technical and expressive qualities of a performance (Bamberger, 1976). Much of the content of piano instruction and pedagogy is concerned with assisting the learner to choose fingerings that will best achieve his or her intended performance. We have extensively reviewed the published literature on fingering, from that of the 18th century to that of the present day (Parncutt, Sloboda, Clarke, & Raekallio, 1997), and have also conducted interviews with expert contemporary pianists (Clarke, Parncutt, Raekallio, & Sloboda, 1997). A broad conclusion from this research is that motor–anatomical and short-term cognitive considerations are foundational in fingering practice and basic-level pedagogy. Influential early authorities (Czerny, 1842; Hummel, 1828; Türk, 1789/1962) concentrated primarily on these considerations rather than on artistic matters. Modern pedagogy still favors the deep grounding of fingering technique on these basic motor–anatomical considerations, from which artistic considerations may, in appropriate circumstances, prompt a departure. Published scale-book fingerings exemplify the centrality of these considerations (e.g., Ridout, 1995).

From a psychological point of view it is perhaps unsurprising that expert pianists place so much importance on the anatomical underpinnings of fingering. The past 200 years have seen a rapid increase in the objective technical difficulty of the piano repertoire (Lehmann & Ericsson, in press). In particular, there has been increasing use of very rapid note sequences (10 or more per second) in the core repertoire. To negotiate such passages effectively requires the optimization of motor–anatomical resources and the elimination of all unnecessary time- and energy-consuming actions. In this respect one can see the tasks facing a pianist as analogous to those facing an athlete attempting to optimize performance beyond previously set limits (Ericsson, Krampe, & Tesch-Römer, 1993). In a real sense, pianists are athletes of the hand.

In order to render operational the principal motor–anatomical constraints in a way that would lead to testable predictions, we developed a model of fingering choice in the form of a computer program (Parncutt et al., 1997). The domain of the model is currently restricted to single-line legato melodies performed with the right hand. Although it will eventually be possible to extend the model to apply to other kinds of music, there is considerable merit in beginning with a manageable but central subset of musical problems. The right-hand melodic line is a key feature of a wide range of music, and much has been written on the fingering issues specific to melodic music. The model incorporates three separate processes. First, it enumerates all anatomically possible fingerings for a series of notes. It does this with reference to a table of maximum and minimum stretches between pairs of fingers (which can be adjusted for the hand size of particular pianists but is currently set to represent an "average" adult hand). Second, the model predicts the overall difficulty of a fingering as a linear sum of contributions from various sources. Third, the model ranks the fingerings in order of calculated difficulty. The fingerings with the lowest overall difficulty scores are predicted to be the fingerings that an expert pianist is likely to choose, other things being equal.

The model currently accounts for 12 sources of difficulty. These are presented in the form of a set of rules, based on motor–anatomical principles such as those set out in the pedagogic writings of Türk (1789/1962). Some of the rules relate to the use of an individual finger (e.g., Fingers 4 and 5 are anatomically weaker than other fingers because of their smaller size and lower agility, and a cost is proposed for any use of these two fingers). Other rules relate to transitions between fingers; for example, it is not optimal to place the fourth finger on a black note after the third finger on a white note because the extensor tendon for the fourth finger is joined to that of the third in such a way that the extension of the fourth finger is inhibited by the flexion of the third, but not vice versa (Green, 1988). A list of the 12 rules is given in Appendix A, along with brief accounts of the rationales underlying them; for more detailed justifications of each rule, see Parncutt et al. (1997).

The model is deliberately simplistic in a number of ways. It takes no account of structural or musical issues and no account of any sequential dependencies involving more than a single note and its immediate temporal neighbors. This is a deliberate tactic we have used in order to see how much of fingering practice can be accounted for by rules at the lowest level of complexity. Only when it is known that these rules cannot account for significant behaviors do we have strong justification for postulating and testing more complex rules. The model is also simplistic in that it weights each rule somewhat arbitrarily (but roughly equally) according to our intuitions as pianists. This is also, we believe, justified as a first approximation. Finally, the model as applied here takes no account of anatomical differences between pianists' hands in terms of size, joint flexibility, and so forth (see Wagner, 1988) but applies to a notional "average male hand" derived from consensus among the four of us, who are all pianists. The collection of anatomical data from pianists in order to adjust weights and spans within the model to better predict individual behavior is an important task, but in this article we focus on randomly constituted groups defined by expertise, rather than on individuals.

Given the existence of such a fingering model it becomes possible to test a number of specific hypotheses that flow from the overarching hypothesis that expertise in piano playing is a result of optimizing motor–anatomical resources. We predict that expert pianists will choose fingerings with lower overall difficulty scores than those chosen by novices. In the current study we define choice as the unpremeditated choice-in-action of sight-reading, rather than the premeditated choices of the prepared performance or written prescription. In Parncutt et al. (1997) we presented some data on considered "best" fingerings offered as written prescriptions and discussed their fit to the fingering model in some detail. It is an open question as to whether experts and novices will differ in their use of all rules or whether there is a specific subset of rules whose use increases particularly significantly as a function of expertise. One possibility is that experts are superior to novices in their

use of rules concerned with minimizing the difficulty of transitions between hand positions, whereas novices equal experts in their use of rules concerned with minimizing difficulty within an essentially static hand position. In this study we addressed this question by estimating conformance to each rule separately.

Although the 12 rules are prima facie candidates for rules adopted by pianists, we do not believe it is feasible that conscious application of these rules takes place in sight-reading of rapid passagework (although such conscious deliberation may often be a part of the determination of a final fingering for performance). It seems much more likely that these rules underlie pattern-recognition procedures that select fingering programs from a lexicon built up from extensive prior experience with the piano repertoire. Such a lexicon may include "standard" fingering patterns prescribed by editors and pedagogues for handling frequently occurring note configurations. There is a considerable body of knowledge that links expertise to pattern matching in other domains (Ericsson & Smith, 1991), and recent research shows that music is no different in this respect. For instance, measures of the time intervals between the onsets of successive notes of equal notated duration have revealed a motor programming system that is deeply sensitive to structural aspects of the music (e.g., Clarke, 1985, 1988; Repp, 1990; Shaffer, 1981; Sloboda, 1983; Todd, 1985), including phrase structure, meter, and harmonic tension. Other investigators (e.g., Palmer & van de Sande, 1993, 1995; Sloboda, 1976, 1985) have used performance errors to reveal how performance is directed by higher order patterns. For instance, many performance errors involve note substitutions that are based on structural equivalence rather than physical closeness (i.e., in the context of a prolonged G-major chord, substituting B for D rather than the physically closer C). Such findings generalize to other performance domains such as language (e.g., Garcia-Albia, del Viso, & Igoa, 1989; Garrett, 1980). These data fit rather well with hierarchical models of musical representation and motor programming, in which high-level groupings and constraints are "unpacked" into increasingly detailed subcommands at the time of performance, rather than performance details being held in a low-level output buffer for long periods of time. We believe that fingering is probably "unpacked" at the time of performance in the same way.

One aspect of expertise that appears in many situations is consistency. Where rules can be shown to exist, it appears that experts apply them more consistently than do novices (for an example in the domain of piano performance see Sloboda, 1985). In line with this we predicted that experts would be more consistent than novices in the fingering solutions they adopted. In the current study we tested for two forms of consistency. Our first hypothesis (consistency across performances) was that experts who sight-read the same piece for a second time would produce more direct replications of the earlier fingerings than would novices. This is because experts have a greater store of learned fingering patterns that are capable of being applied in more circumstances. Our second hypothesis (consistency within performance) was that where equivalent note passages appeared at several points within a piece, experts would finger them more similarly than would novices. This is for the same reason cited in the first hypothesis but also because we expected experts to be less susceptible to "distraction" from suboptimal fingering patterns "forced" on them by different preceding contexts.

A simple example of this is shown in Figure 1. Figure 1 (i) shows a repeating two-note pattern along with a right-hand fingering that is optimal according to the model and pedagogic tradition. Parts (ii) and (iii) show the same pattern preceded by different contexts. The context in (ii) does not affect the fingering of the repeated pattern, because the next-note-next-finger principle is not violated. In (iii), the next-note-next-finger principle is followed, and the fingering of the repeated pattern is changed accordingly. In (iv), the next-note-next-finger principle is violated so that the fingering can return to that shown in (i). In this example (which is deliberately simplistic and presented for the purposes of illustration only), we predict experts will prefer (iv), which overrides the demands of the immediate situation (using adjacent fingers for adjacent notes) for the longer-term advantage of having stronger fingers on the repeated note pair; whereas novices might prefer (iii). This predicted characteristic of experts is consistent with the finding that expert sight-readers look further ahead in reading than do novices (Sloboda, 1974, 1984).

There is one twist to the consistency-within-performance

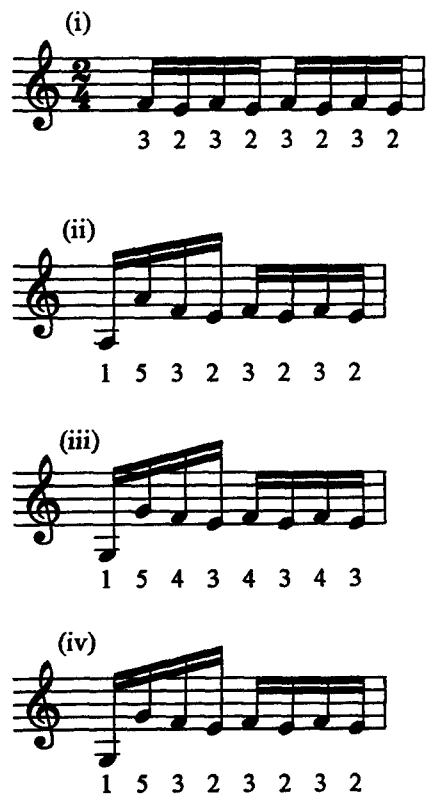

Figure 1. Alternative fingerings for variants of a simple melodic fragment.

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prediction. This is the subprediction that such consistency may break down for experts at the end of two identical passages followed by passages that are different in crucial ways. In such circumstances, we predicted that experts would be more likely to modify the end fingering in order to position themselves more advantageously for what was to follow. This apparently contradictory prediction is based on the assumption that experts look further ahead than do novices.

Method

Participants

Participants were 16 pianists ranging from established professional performers through conservatory-level students to beginners. Beginners were defined as individuals of at least conservatory level on an instrument other than the piano who had been learning the piano for 3 years or less. This definition was used to ensure that any differences between these and other pianists were not due to lower levels of musical literacy or understanding. Participants' actual levels of expertise were estimated from two pieces of biographical data: the number of years that the individual had been playing the piano (range = 3–48) and the number of public concerts given in the preceding 12 months (range = 0–60). These two numbers were summed to give a crude experience rating that ranged from 3 to 80. For some analyses the participants were divided into three groups: masters (n = 6, experience range = 51–80), experts (n = 5, experience range = 32–37), and novices (n = 5, experience range = 3–21).

Experimental Materials

Materials were seven pieces from Carl Czerny's *160 kurze Übungen* [160 Eight-Bar Exercises] *Opus 821*. Although these pieces were originally published in the mid-nineteenth century, in this study we used the 1920 Peters edition (Czerny, 1845?/1920). The pieces were No. 1 in C Major (Piece A), No. 37 in A Major (Piece B), No. 38 in A Major (Piece C), No. 54 in C# Major (Piece D), No. 62 in D Minor (Piece E), No. 66 in G Minor (Piece F), and No. 96 in B Minor (Piece G). This particular collection has a number of advantages for the present study. Each piece is short but presents repeated opportunities for a pianist to address a particular technical or fingering issue likely to recur in core classical piano repertoire. Because the work was written with purely technical ends in mind, there are few interpretative markings; nonetheless each piece is a real, if simple, musical entity that requires some basic expressive shaping and is not a mere technical exercise. Finally, these pieces are somewhat unfashionable currently and not well-known by contemporary pianists. The seven specific pieces were chosen from the 160 exercises to fulfill two conditions: first, that each piece contain a predominantly continuous legato 16th-note single-line melody in the right hand; second, that each piece be based around a different type of melodic figuration posing unique fingering problems. The scores of the pieces are presented in Appendix B, with Czerny's original fingering included. It should be noted that not every note is assigned a finger by Czerny. Generally, unassigned fingerings are those considered by the composer or editor to be obvious, although in reality there are often different options for unfingered notes. This phenomenon is of some interest in its own right, as a pointer toward rules and common patterns within a pianistic culture. The participants were presented with versions of these pieces from which the composer's intended fingerings had been erased.

Data Collection Environment

Participants performed on music instrument digital interface (MIDI) keyboards, either a Yamaha Disklavier (a real piano with MIDI output) or a Yamaha Clavinova (one of the top-range synthesised touch-sensitive MIDI keyboards which reproduces piano tone and touch to an acceptable degree). Two of the authors who were practicing pianists judged that the difference in force characteristics of the two keyboards was barely perceptible and unlikely to significantly influence fingering choices. Performance data (onset time, velocity, and offset time for each note) were stored on an Apple Mac Powerbook 150 in MIDI form for later analysis. Simultaneously, a video recording with sound was made of the performance with a JVC camera positioned approximately 1.5 m vertically above the keyboard, which allowed a bird's-eye view of the pianist's right hand and, in particular, allowed determination of the finger used to play each note. Occlusions of the hand by the head or upper torso never occurred, although sometimes the thumb was occluded by other fingers. This did not pose a problem for transcription because it was always possible to directly observe the location of the other four fingers and the keys, if any, that they were depressing. Any key not being depressed by a visible finger must therefore have been depressed by the thumb. The camera was connected to a Panasonic VHS recorder.

Testing rooms in two different universities were used for data collection. Each room was in a quiet location but with a natural room acoustic (i.e., no soundproofing or acoustic tiling).

Procedure

This study formed part of a longer session with each participant that lasted approximately 1 hr and included an interview and other tasks unrelated to this study. After some other tasks that familiarized the participant with the instrument and the testing environment, each of the seven Czerny pieces was presented sequentially for sight-reading. The participant was instructed to view each piece for a few seconds and then give a performance of the right-hand melody at sight. Each participant was asked to choose a performance speed that would enable predominantly accurate performance. After each piece had been played once, the participant played an unrelated piano piece at sight, which acted as a distractor. Then the participant repeated the sight-reading task with the same seven melodies in the same order. No instructions were given to participants regarding fingering.

Results

Consistency Between Performances

A prerequisite for the analysis of fingering consistency was the identification of note errors. We did this by comparing the two takes of each piece and deleting notes from one or the other take until the sequence of notes in each take was identical. All deleted notes were counted as errors. We deliberately avoided comparing the performance with the score, anticipating that some pianists may have misread the score in the same way on both occasions and that comparison with the score might have resulted in unnecessary elimination of data. Given the inherent ambiguity of the error-detection process, we developed two contrasting algorithms for error detection and compared the results obtained with each. We call them the recursive and nonrecursive algorithms (see Appendix C). We compared the number of

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notes deleted by the two algorithms from the performances of each pianist and found that there was a moderate positive correlation between the two measures of .60 ($p < .05$; Pearson's product–moment for this and all other reported correlations and two-tailed significance test here and in all other analyses). We decided to proceed using the data from the nonrecursive algorithm, which yielded lower error estimates for all pianists (the mean difference between the two algorithms was 11.4%). This allowed the retention of the greatest amount of usable data for the consistency analysis (94% as opposed to 83%). For each pianist we computed a total error percentage score by summing all note inconsistencies between takes (omissions, substitutions, order reversals, or additions). This percentage ranged from 1.6% for the best pianist to 16.9% for the worst. Mean error rates for the three expertise groups were 9.4% for novices, 4.9% for experts, and 4.1% for masters.

For those notes identified as correct (by virtue of being the same in both performances) a second program determined whether the finger used was the same or not and computed a mean percentage fingering consistency score for each pianist. This was the percentage of notes played with the same finger in both performances. The program also calculated each pianist's mean speed of performance (in notes per second) by dividing the total playing duration by the total number of notes played, including errors). The correlation between the pianists' expertise and their mean consistency was positive and highly significant ($r = .66$, $p < .01$), as was a one-way analysis of variance between the three expertise groups, $F(2, 13) = 33.9$, $p < .001$. Mean overall consistency was 81% (masters = 85%, experts = 85%, novices = 74%). Post hoc Tukey tests showed that masters and experts were significantly more consistent than novices. Even novices, however, achieved a reasonably high level of consistency. This was achieved partly at the expense of speed. Across the 16 pianists, average speed correlated significantly with experience ($r = .65$, $p < .01$) such that greater experience was associated with greater speed. Average speed ranged from 1.3 notes/s for the slowest player to 7.9 notes/s for the fastest player. Table 1 shows average speed, consistency, and error for each individual pianist.

We investigated the relationship of fingering consistency to speed and accuracy of performance using the data in Table 1. The correlation between consistency and errors was −.61 ($p < .01$), which showed that the greater the fingering consistency, the fewer the errors. The correlation between experience and errors was −.54 ($p < .05$), which showed that the greater the expertise, the fewer the errors. This is consistent with the notion that performance accuracy is dependent on the existence of readily available (and thus reproducible) fingering strategies. There was an insignificant correlation between speed and error of −.43. Thus, there was no strong evidence of a speed–error trade-off. Speed and consistency were, however, highly correlated ($r = .87$, $p < .001$), which showed that faster players were more consistent in their fingering. It is reasonable to suppose that the existence of consistent fingering plans for constructing integrated motor programs is one of the main preconditions for high-speed performance.

Table 1
Mean Speed, Consistency, and Error for 16 Pianists

Category	Experience rating	Speed (in notes per second)	Consistency (%)	Error rate (%)
Novice	3	1.3	63.6	16.8
Novice	7	2.4	69.2	8.9
Novice	13	3.4	71.9	14.1
Novice	19	3.5	76.2	1.6
Novice	21	5.0	83.4	5.8
Expert	32	5.5	78.9	6.0
Expert	34	3.8	81.0	1.7
Expert	35	3.7	88.4	4.0
Expert	36	6.5	88.8	2.1
Expert	37	7.6	90.8	10.7
Master	51	4.2	77.5	3.9
Master	55	5.0	84.6	3.9
Master	63	4.6	81.4	4.9
Master	68	5.9	82.2	5.6
Master	75	5.2	83.9	5.0
Master	80	7.9	93.4	1.7

Fingering Rule Use

We further analyzed the relationship of fingering strategies to consistency, error, and experience by applying some predictions of Parncutt et al.'s (1997) fingering model to the data obtained. We gave the fingering pattern chosen for the first 100 notes of each performance a difficulty rating by summing the difficulty scores obtained by applying each of 12 fingering rules specified in the fingering model. The smaller any such score, the "easier" the fingering chosen, according to the model. Although some pieces were longer than 100 notes, taking the same number of notes from each piece allowed direct comparison of difficulty between pieces in all subsequent analyses. Because each piece was consistent in patterning throughout its length, we do not believe that the omission of some end passages significantly affected the pattern of results. It should be noted that for all analyses of rule use, performance errors were included and the note-deletion program used for the consistency analyses was not applied. Thus, the rules were applied to the fingerings chosen for the notes actually played, whether these notes were correct or not.

A three-way analysis of variance was carried out on difficulty scores for each individual rule and for all rules summed, with take (first or second performance), group (master, expert, and novice), and piece (A through G) as independent variables. Neither the main effect of take nor interactions including this variable were significant in any case. This suggests that the sight-reading environment did not allow any significant improvement in fingering to be achieved between the first and second takes.

The main effect of group on the total of all rules was significant, as shown in Table 2. A post hoc Tukey test showed that experts chose significantly "easier" fingerings than did masters or novices, who did not differ significantly from one another. Analysis of individual rules showed that not all rules contributed in the same way to this effect. Table

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2 shows that Rules 1, 3, 7, 8, 9, and 11 did not significantly differentiate the groups. Rules 4 and 5 showed a U-shaped function, with experts choosing easier fingerings than both masters and novices. Rules 2 and 6 showed a linear effect, with masters choosing the most difficult fingerings and novices choosing the easiest. Rules 10 and 12 showed the opposite linear effect, with masters choosing the easiest fingerings and novices choosing the most difficult.

It is of note that around 90% of the difficulty values are associated with five rules (Rules 2–6), although this is partly a function of the intuitive weightings assigned to different rules within the model. It appears that in these pieces pianists of all levels generally avoid fingerings that create difficulty on the other seven rules. Although pianists of differing expertise show slightly different profiles, the overall picture revealed by Table 2 is one of considerable similarity between pianists. In only one case (Rule 10) is the score of the highest scoring group more than double that of the lowest scoring group. The results from Rule 10 show that novices are far more likely to place a thumb on a black key than are masters or experts.

There were also several significant main effects for piece ($df = 6, 182$ in all cases). These were complicated in some cases by two-way interactions of piece and group ($df = 12, 182$ in all cases). In light of the small accumulations of scores for 7 of the 12 rules, as revealed in Table 2, only those rules with significant accumulations are shown in Table 3. Total mean difficulty scores ranged from 137 (Piece G) to 377 (Piece D). Pieces D, E, and F were close in overall difficulty scores and accounted between them for the highest difficulty scores on each individual rule. If difficulty ratings (produced by the model) correspond to actual played difficulty, then we should expect pianists to make most performance errors on those pieces with the highest difficulty ratings.

Mean percentage error on each piece is shown at the bottom of Table 3. The correlation between these seven pairs of means is .63. Because the number of pairs was too small for this correlation to achieve significance (a correlation of .70 would be needed to achieve significance at the .05 level), we calculated a correlation of error with total difficulty using as data each of the 224 performances (7 pieces × 2 takes × 16 performers). This yielded a significant positive correlation of .21 ($p < .001$). Difficulty as assessed by the model is thus a modest predictor of performance accuracy. It is also a significant but less good predictor of consistency ($r = -.17$, $p < .01$). Looking at rules individually across the 224 performances, we find that these correlations are mainly attributable to Rules 4 and 5 (significantly correlating with error at .21 and .23, respectively). More frequent and larger hand-position changes reduce both accuracy and consistency. It is clear, however, that the predictive power of these rules is low (accounting for at most 4% of variance) and that a full account of performance error must take into account many other factors.

Of the significant piece by group interactions indicated in Table 3, only one achieves significance beyond the .01 level. This is the effect for the small-span rule (Rule 2). The interaction is shown in Figure 2. Figure 2 shows that in Pieces D, E, and F, experts (and to a greater degree masters) were more prepared than novices to play fingerings that were difficult by virtue of involving smaller intervals between fingers than a comfortable span. Masters and experts abandoned the next-note-next-finger principle in favor of "squashed" hand positions. The piece with the most extreme differences between experts and novices was Piece D. If one looks at all other rules in this piece, it is possible to see whether this "sacrifice" of the small-span rule allowed compensating gains on any of the other rules. Table 4 shows the mean difficulty score attributed by each rule to the three expertise groups on this piece. One-way analysis of variance shows that the only substantial significant gain for masters and experts over novices was on Rule 10 (thumb on black; $df = 2, 28$ in all cases). The more expert players avoided placing the thumb on black, particularly when the preceding or following note was a white key.

Table 2
Mean Difficulty of Fingerings Chosen by Masters, Experts, and Novices According to the Predictions of the Model

Rule number and name	Masters	Experts	Novices	$F(2, 182)$	p
1. Stretch	5.8	5.5	6.0	0.19	*ns*
2. Small span	67	60	54	8.69	<.001
3. Large span	19	18	20	0.15	*ns*
4. Position change count	43	38	45	5.26	<.01
5. Position change size	77	69	77	5.23	<.01
6. Weak finger	27	26	23	5.29	<.01
7. Three-four-five	4.5	4.6	3.6	1.87	*ns*
8. Three to four	3.9	3.7	4.0	0.19	*ns*
9. Four on black	1.2	1.2	1.5	0.20	*ns*
10. Thumb on black	5.9	8.1	13	17.69	<.001
11. Five on black	2.6	2.5	2.0	2.42	*ns*
12. Thumb passing	2.8	3.2	4.3	4.96	<.01
All rules	260	241	254	5.05	<.01

Note. F ratio is for the main effect of group within a three-way analysis of variance of take (first or second) by piece (A through G) by group, with participants within groups as the error term.

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Table 3
Mean Difficulty (Rounded) of Fingerings in Each of Two Performances of Seven Pieces According to Five Rules From the Model, Compared With Mean Percentage Error and Between-Performances Consistency for the Same Pieces

Rule number and name	Piece							$F(6, 182)$	p	Interaction
	A	B	C	D	E	F	G			
2. Small span	43	20	71	106	71	55	59	57.4	<.001	yes
3. Large span	2	14	21	16	23	54	3	210.9	<.001	yes
4. Position change count	39	27	30	71	57	58	14	79.0	<.001	no
5. Position change size	44	47	62	110	123	116	21	220.0	<.001	no
6. Weak finger	28	29	22	29	26	30	17	19.7	<.001	no
All rules	170	158	233	377	331	359	137	285.2	<.001	yes
Error (%)	2.4	5.1	4.0	8.0	10.4	6.4	6.4			
Consistency (%)	90	87	90	81	79	82	82			

Note. F ratio is for the main effect of piece within a three-way analysis of variance of take (first or second) by group (masters, experts, novices) by piece, with participants within groups as the error term. Interaction refers to the presence of a significant interaction between piece and group (master, expert, novice; $df = 12, 182$).

The data from the individual fingering rules suggest strongly that the relationship between expertise and rule use is complex. Some rules yield increases in difficulty from novices to masters, others yield decreases, and yet others show either no change or a U-shaped function. To see if a simpler picture might be obtained by combining similarly behaving rules, we carried out a principal-components factor analysis with varimax rotation on the scores derived from the 12 fingering rules, for all pieces and takes of each participant. This yielded a five-factor solution, with the five factors accounting for 77.8% of total variance. Factor 1 (high positive loadings on Rules 1, 3, 10, and 11) accounted for 33% of the variance. This may be seen as a Stretch factor, because all the rules loading high on this factor involve placing fingers beyond their comfortable range. Factor 2 (high positive loadings on Rules 2, 4, and 5) accounted for 15% of the variance. This may be seen as a Position Change factor, because all rules loading high on this factor precipitate, or make possible, a change in the hand position. Factor 3 (high positive loadings on Rules 6, 7, and 8) accounted for 12% of the variance. This may be seen as an Agility factor, because all rules loading high on this factor include transitions involving the weak fourth and fifth fingers. Factors 4 and 5 loaded on only one item each (Rules 9 and 12, respectively) and each accounted for 9% of the variance. We obtained a difficulty rating for each performance on each factor by summing the ratings of the high-loading rules for that factor. These ratings were entered into one-way analyses

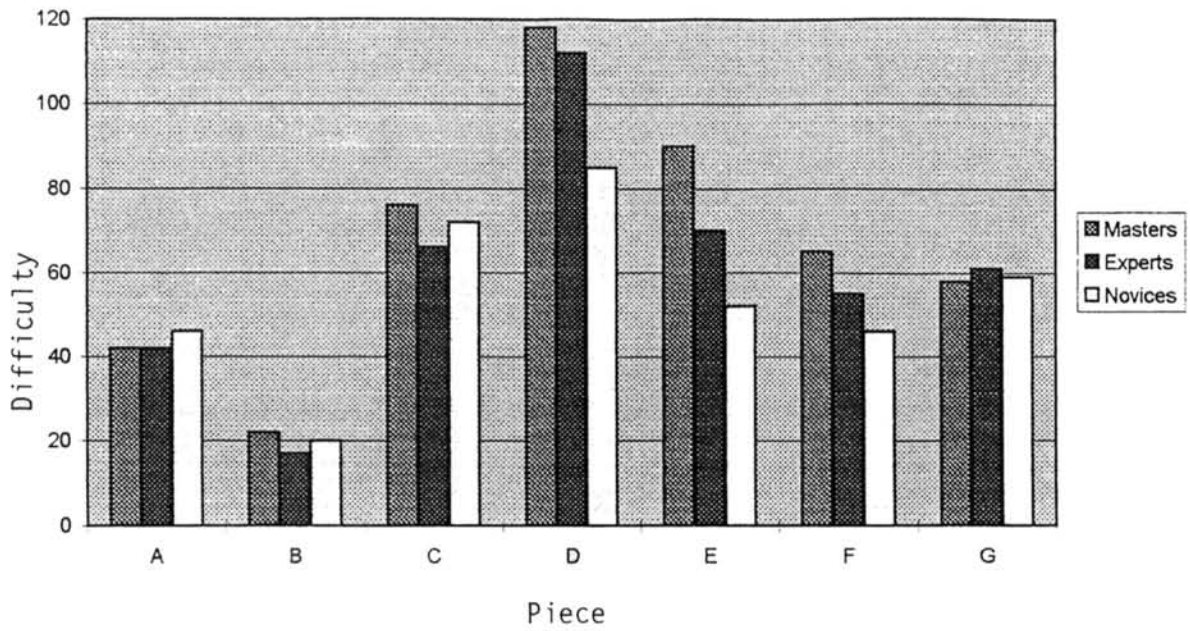

Figure 2. Mean difficulty ratings of fingerings in each of seven pieces (A through G) according to the small-span rule (Rule 2) for masters, experts, and novices.

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Table 4
Mean Difficulty (Rounded) of Fingerings on Piece D According to the Predictions of the Model, for Masters, Experts, and Novices

Rule number and name	Masters	Experts	Novices	$F(2, 28)$	p
1. Stretch	10	10	10	0.05	*ns*
2. Small span	118	112	85	4.31	<.025
3. Large span	15	14	19	2.85	*ns*
4. Position change count	76	65	70	1.06	*ns*
5. Position change size	116	107	105	1.82	*ns*
6. Weak finger	31	31	24	6.15	<.01
7. Three-four-five	4	6	3	3.11	<.06
8. Three to four	1	2	2	0.41	*ns*
9. Four on black	0	0	1	12.40	<.0001
10. Thumb on black	14	17	31	7.18	<.005
11. Five on black	4	8	2	7.57	<.002
12. Thumb passing	3	4	5	1.98	*ns*
All rules	394	377	358		

Note. F ratio is for the main effect of piece within a three-way analysis of variance (first or second) by group (masters, experts, novices) by piece, with participants within groups as the error term.

of variance with group as the independent variable. Only Factors 3 and 5 showed significant differences between groups, with masters obtaining the highest difficulty ratings on Factor 3, $F(2, 217) = 3.20$, $p < .05$, and the lowest ratings on Factor 5 (see Table 5). It appears that the more expert pianists are more prepared to use weak fingers, which attract high difficulty ratings on the model, but are less prepared to use thumb passes, which also attract high difficulty ratings. That these sets of rules should be in opposition makes sense. If a pianist is not prepared to use the weak Fingers 4 and 5 very much, then keeping rising or falling passages legato will require more thumb passing (e.g., playing an eight-note scale while avoiding Finger 4 will require two thumb passes [1231231], whereas using Finger 4 will require only one pass [1234123]).

In sum, the analysis of rule conformance suggests that there is not a simple decrease in the difficulty of fingerings chosen as expertise increases. It may be that there is no fingering that can have low difficulty on all rules for these pieces. Instead, masters, experts and novices differ in the way they trade one rule against another. Novices prefer to keep the fingers near their optimal separation, neither too stretched nor too squashed, which suggests a conformance to scale-book patterns where possible ("next-note-next-finger"). Experts and masters are more prepared to squash fingers close together, particularly if by doing so they can avoid putting the thumb on a black note. Experts and masters are also more willing than novices to use finger patterns involving movements between the third, fourth, and fifth fingers. Novices, by avoiding these fingers, have to resort to more passing under of the thumb. Experts prefer fingerings that avoid hand-position changes to a greater extent than do either novices or masters. Novices may have to change hand position more often to avoid overuse of the weak fingers. Masters may be more concerned with ergonomically optimal fingering, such as avoiding thumb on black, than with maintaining constant hand position. Finally, rule difficulty, particularly of hand-position changes (Rules 4 and 5), statistically predicts both error and consistency in performance.

Consistency Within a Single Performance

A computer program was written that scanned a file containing a representation of the entire score of each of the seven pieces (right-hand melody only). The scanning process identified any melodic fragment of seven notes or more that was reproduced anywhere else in the entire score. It

Table 5
Mean Factor Difficulty Ratings Chosen by Masters, Experts, and Novices According to the Predictions of the Model

Factor	Masters	Experts	Novices	$F(2, 221)$	p
1. Stretch	33	34	41	1.75	*ns*
2. Position Change	187	167	176	1.12	*ns*
3. Agility	35	35	31	3.19	<.05
4. Rule 9	1.2	1.2	1.5	0.20	*ns*
5. Rule 12	2.8	3.2	4.3	4.96	<.01

Note. F ratio is for the main effect of piece within a three-way analysis of variance (first or second) by group (masters, experts, novices) by piece, with participants within groups as the error term.

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identified not only exact note-pattern replications but also any pattern that was a replication from the point of view of its spatial configuration on the keyboard (which as a result should, all other things being equal, be fingered in the same way). Operationally this was defined as any pattern that shared both the same set of intervals between successive notes and also the same configuration of black and white keys, regardless of octave or chroma. So, for instance, the sequence C–Eb–E would be rated as equivalent to G–Bb–B, but not to D–F–F#. The minimum length of seven notes for repeating sequences was determined on pragmatic grounds. The smaller the fragment the more often it recurred, and very often such small fragments were embedded within larger fragments that the program had already identified. Setting a minimum of seven notes avoided significant double counting and made the analysis manageable. The operation of this program yielded 23 sequences (which we will call repeat fragments) that recurred at least once, at a distance of 12 notes or more from the original. Most such pairs were within a single piece (as would be expected from the fact that each piece repeated similar sequences, which were chosen to be different from those found in other pieces). Two pairs, however, spanned different pieces. The sequences ranged in length from 7 notes to 19 notes. Because each participant performed the sequence twice, fingering data exist for 32 performances of each pair (736 performance pairs in total). Nineteen of these pairs (2.5%) contained note errors, which decreased the number of pairs available for analysis to 717.

We estimated overall fingering consistency between the repeat sequences by counting the number of notes for which the same finger was used in each repeat. A one-way analysis of variance showed that expertise groups differed significantly on this measure, $F(2, 714) = 14.8$, $p < .0001$ (see Table 6). Post hoc Tukey tests showed that masters and experts were significantly more consistent than novices but not different from one another. Between-performances consistency for these fragments is also shown in Table 6. It should be noted that within-performance consistency was on average around 23% lower than between-performances consistency measures, for all groups. The different contexts of the two repeat sequences are likely to be responsible for this drop.

We measured overall consistency at the end of each repeat sequence in two ways, by analyzing data from only the last four or only the last two notes of each sequence. In each case, a significant group difference was observed: one-way analysis of variance, $F(2, 714) = 10.6$, $p < .0001$, and $F(2, 714) = 6.29$, $p < .0025$, respectively; see Table 6. Post hoc Tukey tests showed that masters were significantly more consistent than novices in both cases but that experts were only significantly more consistent than novices in the case of the last four notes. There is no evidence that either masters or novices drop in overall consistency toward the end of a repeat sequence. If anything, average consistency seems to rise toward the end of these sequences.

However, the prior analysis did not take into account the degree of difference between the immediately following contexts. It could be argued that the following context would only be likely to encourage different end fingerings if the amount and direction of movement were substantially different between the two contexts. Therefore it would be important to relate fingering consistency to the degree of subsequent difference in continuation. A simple and appropriate measure of difference is pitch difference. Accordingly, we computed the pitch difference in semitones between each of the three notes immediately following the final note of the repeat sequence. So, for instance, if the first note after the first occurrence of a same-pitch repeat sequence was D4 and the first note after the second occurrence was G4, then the difference between them would be 5 semitones. These differences, $d1$, $d2$, and $d3$, could also be summed to generate differences $d(1 + 2)$ and $d(1 + 2 + 3)$. Figure 3 gives examples of repeat sequences with the smallest (1 semitone) and largest (32 semitones) values of $d(1 + 2 + 3)$.

Table 6
Mean Fingering Consistency (%) for Repeat Fragments

Consistency measure	Masters	Experts	Novices	Total
Within-performance				
Whole sequence	63.9	61.9	49.4	58.8
Last four notes only	65.8	64.7	51.5	61.1
Last two notes only	64.6	58.0	51.1	58.4
Between-performances				
Whole sequence	85.9	86.1	72.3	81.7

Correlations were computed between the three consistency measures (all notes, last four notes, and last two notes) and the five following-context difference measures ($d1$, $d2$, $d3$, $d(1 + 2)$, and $d(1 + 2 + 3)$. This was done for each expertise group separately. In no case did either the all-notes or the last-four-notes consistency measure correlate significantly with any of the following-context measures. This suggests that there are no artifactual relationships (of general difficulty, for instance) between the repeat sequences and their contexts. There were, however, a number of significant negative correlations between the last-two-notes consistency measure and the following-context difference measures. The correlations provide some evidence to support the prediction that pianists are likely to be less consistent in fingering at the very end of a repeat sequence in direct proportion to the degree of pitch difference in the following contexts.

To explore these effects further, we carried out two-way analyses of variance using last-two-notes scores as the dependent variable. The factors were expertise group (master, expert, novice) and degree of difference ($d[1 + 2 + 3]$; three levels—low [0–4 semitones], medium [5–11 semitones], and high [12 semitones or over]). The main effects of both group, $F(2, 708) = 6.48$, $p < .0025$, and degree of difference, $F(2, 708) = 9.54$, $p < .001$, were significant, but the interaction between these two factors was not, $F(4, 708) = 0.716$. The means for the interaction effect are shown in Table 7. These means suggest that the reduction of consistency in the last two notes according to following-context difference is not expertise dependent. All three expertise groups showed fingering that was sensitive to the following context. When the following context differed by 12 or more

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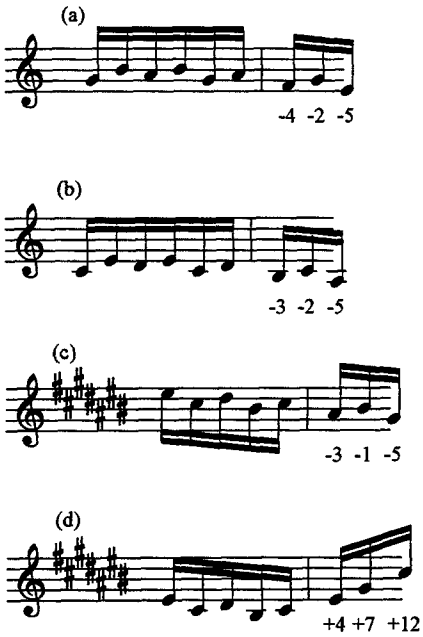

Figure 3. The pairs (a)–(b) and (c)–(d) show the concluding notes of two repeat sequences, with the first three notes of the continuations sequence after the vertical line. The numbers represent the distance in semitones of each note from the last note of the repeat sequence. The pitch difference between continuations is obtained by subtracting the values under (a) or (c) from the corresponding values in (b) or (d). (a)–(b) shows the smallest pitch difference in the stimulus set; (c)–(d), the largest.

semitones summed across the three following notes, then fingering consistency in the last two notes of repeat sequences was significantly lowered for all pianists.

Discussion

These data strongly support the notion that much of fingering choice in fluent sight-reading can be accounted for by the requirement to reduce certain ergonomically unfavorable actions to a minimum. In these pieces, no pianist or any level of expertise was willing to frequently use weak finger transitions involving the third, fourth, and fifth fingers. Similarly, all pianists seemed to predominantly avoid placing the fourth or fifth finger on a black note and passing the thumb under onto a black key. Our data therefore confirm that pianists do, by and large, conform to the prescriptions of the classic fingering literature.

Table 7
Mean Percentage Consistency as a Function of Expertise and Following-Context Pitch Difference

Following-context pitch difference	Masters	Experts	Novices	All
Low (0–4 semitones)	71.7	64.0	55.1	64.1
Medium (5–11 semitones)	66.4	66.9	56.3	63.4
High (12+ semitones)	58.7	45.4	43.5	49.8

The prediction that expert pianists would choose fingerings of lower overall difficulty than would novices was only partially supported. Rather, pianists of different expertise levels tend to adopt a different trade-off between conflicting rules. Our tentative prediction was that pianists of greater expertise would avoid difficult hand-position changes. Although this holds when one compares experts and novices, masters seem to "revert" to the same level of difficulty as novices. They seem to tolerate hand-position changes rather than use ergonomically weak maneuvers such as putting a thumb on a black note. On the other hand, maybe as a result of finger strengthening through large amounts of practice (or through better use of the arm to compensate for these fingers' inherent weakness), masters and experts are somewhat more willing than novices to use transitions from the third finger to the weaker fourth and fifth fingers. This may be particularly important in contemporary piano repertoire, where the writing style may demand more equal treatment of all the fingers. Indeed, we may speculate that experts and masters were more likely to have had significant exposure to contemporary repertoire than novices, which thus may have allowed them to develop nonstandard fingerings. The present study does not, however, allow us to test this speculation because data on pianists' repertoires were not collected.

The most striking difference between novices and the other pianists is the novices' relative unwillingness to use stretched or squashed finger transitions. Novices like to keep fingers the standard distance apart that will allow the standard scale-book pattern of finger–note mapping (next-note-next-finger). Masters and experts are more flexible in their use of nonstandard spans, probably in the interests of ergonomically strong outcomes, such as keeping the thumb on white notes. The capacity to make frequent use of wide stretches may be one characteristic of the well-trained pianistic hand.

A general observation about the rule-use data is that significant group differences occur in the use of rules (such as Rule 12 [Factor 5]) that are not broken to any great degree by pianists of any level of expertise (in the particular case by rarely or never passing the thumb under onto a black note) and that thus yield very low mean difficulty values. This is because error terms are also very low in such cases, significantly lower than those for some rules where difficulty levels and mean group differences in those levels are much higher. But low difficulty scores do not imply that these rules are unimportant for pianists. Rather, it is a sign that these rules are fundamental and are therefore rarely broken. A closer analysis of those contexts in which novices are "forced" into breaking such rules will be an important component of future research.

The third prediction was that higher levels of expertise would be related to higher levels of consistency between consecutive performances of the same piece. This was shown to be the case, although even the novices achieved a relatively high average between-performances consistency of 74%. Consistency was negatively, but weakly, correlated with the difficulty of the fingerings chosen, particularly for the rules involving changes of hand position. Pianists with lower consistency made more errors, and the pieces with

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high fingering difficulty also elicited more errors. These data suggest that more consistent adherence to rules is, indeed, partly responsible for increased between-performances consistency and accuracy. However, the fact that overall rule use was not strikingly different for pianists of different levels of expertise suggests that another factor must be accounting for some of the consistency effect. Typically a large number of fingerings meet the requirements of any particular fingering rule. Pianists need more than the rules to determine an actual fingering to play. It seems to us most likely that expert pianists have stored a set of specific preferred fingering solutions for note sequences that frequently occur in particular musical contexts. These Czerny excerpts contain prototypical, if rudimentary, examples of particular musical devices commonly found in the classical piano repertoire (e.g., Haydn, Mozart, Beethoven). The solutions learned for these prototypes, although conforming to the rules, cannot be uniquely predicted by the rules. Parncutt et al. (1997) have shown that there are some fingering patterns that are assigned a low difficulty value by the fingering model but that no pianists actually nominate as plausible. These unnominated fingerings may be those that are not prescribed by any tradition or that have some cognitive or figural property that we have not yet considered.

The fourth prediction, that higher expertise would yield higher consistency for repeats of note fragments within pieces, was also upheld. Masters and experts were more consistent than novices, although within-performance consistency was around 20% lower than between-performances consistency for all groups. We argue that this drop can be explained by contextual differences between repeat sequences.

The fifth prediction, that pianists would be sensitive to following context, was also upheld. Pianists were less consistent in their fingering of the last two notes of a repeat fragment in those cases where the next three notes were significantly different in average pitch level. The hypothesis was, however, incorrect in proposing that this effect would occur only for experts. The same effect was found for every expertise level in this study. All pianists seem able to preview large pitch movements and to make consequent adjustments in final fingering to anticipate these. In that respect, even novices are "expert." We propose that this lack of group difference is explained in part by the fact that the novices were actually expert on some other instrument. Their reading skills were thus equivalent to those of the experts, and we may assume in particular that all groups previewed the score to the same extent. In addition, the generally slower performance speeds adopted by novices would have assisted the maintenance of preview. The behavior of all groups may be reflected in their shared possession of a rather crude heuristic that specifies that large pitch changes require hand-position changes.

A significant feature of this study is the use of real, if relatively simple, pieces of composed music as a test bed for our hypotheses. This allows us to conclude, without any extrapolation from more artificial materials, that the principles enumerated by Türk (1789/1962) and followers are significant constraints on the fingering choices made by pianists in the high-demand situation of rapid sight-reading.

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Appendix A

Rules Specified in Parncutt, Sloboda, Clarke, and Raekallio's (1997) Model of Fingering Choice

Definitions

Note

The unit of distance is a semitone, rather than a physical distance. The model ignores small differences in the physical size of intervals spanning the same number of semitones.

Maximum Possible Span (MaxPoss)

The furthest distance between two fingers that can be stretched within the limits of the hand's anatomy. This includes stretches that are uncomfortable or painful and that cannot be sustained in regular performance. MaxPoss is not estimated or used in this model.

Maximum Practical Span (MaxPrac)

The maximum stretch between two fingers that a pianist would actually use to play two simultaneous or legato successive notes. It is estimated by consensus among Parncutt et al. (1997) to represent an average male hand. No fingering is allowed in the model that incorporates stretches larger than MaxPrac.

Minimum Practical Span (MinPrac)

The minimum stretch between two fingers that a pianist would actually use to play two simultaneous or legato successive notes. For Finger pairs 1–2, 1–3, and 1–4, MinPrac is negative, to reflect the passing of the thumb under Fingers 2, 3, and 4. For finger pairs not including the thumb, MinPrac is small and positive (usually, 1 semitone).

Maximum Comfortable Span (MaxComf)

The maximum stretch between two fingers that may be played without noticeable effort. It is defined as two semitones smaller than MaxPrac.

Minimum Comfortable Span (MinComf)

The minimum stretch between two fingers that may be played without noticeable effort. For finger pairs including the thumb, it is defined as two semitones smaller than MinPrac. For other finger pairs, it is set equal to MinPrac.

Maximum Relaxed Span (MaxRel)

The maximum span between two fingers when the hand is completely relaxed. For finger pairs not including the thumb, it is defined as twice the difference between finger numbers in semitones (e.g., MaxRel for Finger Pair 2–5 = 6 semitones). For finger pairs including the thumb, it lies roughly midway between MaxComf and MinRel (see below).

Minimum Relaxed Span (MinRel)

The minimum span between two fingers when the hand is completely relaxed. For all finger pairs, it is defined as one less than twice the difference between finger numbers in semitones (e.g., MinRel for Finger Pair 1–4 = 5 semitones).

Rules

1. *Stretch rule: Add 2 points to the estimated difficulty for each semitone that an interval exceeds MaxComf.*

This rule accounts for the difficulty associated with stretching from MaxComf to MaxPrac spans and from MinComf to MinPrac when passing the thumb under another finger. Example: For D4–C5 (10 semitones) played 2–5, the stretch rule assigns 4 points, because MaxComf for Fingers 2–5 is 8 semitones, 10 semitones is 2 semitones larger, and $2 \times 2 = 4$.

2. *Small-span rule: For finger pairs including the thumb, add 1 point to the estimated difficulty for each semitone that an interval is*

less than MinRel. For finger pairs not including the thumb, add 2 points per semitone.

Pianists tend to prefer fingerings in which neighboring fingers map to neighboring notes in a diatonic scale. We refer to this as the "next-note-next-finger" principle. The small-span rule applies whenever the principle is violated in the direction of smaller intervals, which makes the hand feel cramped. This includes passing the thumb under another finger, or a finger over the thumb. Example: For E4–G4 (3 semitones) played 3–1 (passing the thumb), the small-span rule assigns 6 points, because MinRel for 1–3 (without passing the thumb) is 3 semitones, and passing 1 under 3 through an interval of 3 semitones involves a span that is smaller than MinRel by 3 + 3 = 6 semitones. It is easier to move the thumb sideways relative to the other fingers than to move the fingers sideways relative to each other; the small-span rule consequently assigns a larger penalty to intervals not involving the thumb. Example: For C4–D4 (2 semitones) played 2–5, the small-span rule assigns 6 points, because MinRel between Fingers 2 and 5 is 5 semitones—3 semitones larger—and 3 × 2 = 6.

3. *Large-span rule: For finger pairs including the thumb, add 1 point to the estimated difficulty for each semitone that an interval exceeds MaxRel. For finger pairs not including the thumb, add 2 points per semitone.*

The large-span rule applies whenever the next-note-next-finger principle is violated in the direction of larger intervals. For spans that also exceed MaxComf, the difficulty is further augmented by the stretch rule. Example: For C4–B4 (11 semitones) played 1–3, the large-span rule assigns 4 points, because MaxRel = 7 semitones; and the stretch rule assigns 2 additional points, because 11 semitones is 1 semitone larger than MaxComf for 1–3. Like the small-span rule, and for the same reasons, the large-span rule assigns a larger penalty to intervals not involving the thumb. Example: For C4–G4 (7 semitones) played 2–4, the large-span rule assigns 6 points, because MaxRel = 4 semitones, and 2 × (7 − 4) = 6; and the stretch rule assigns an additional 4 points, because 7 semitones is 2 semitones larger than MaxComf for 2–4.

4. *Position-change-count rule: Add 2 points to the estimated difficulty for every full change of hand position and 1 point for every half change. A change of hand position occurs whenever the first and third notes in a consecutive group of three span an interval that is greater than MaxComf or less than MinComf for the corresponding fingers. In a full change, the second finger is the thumb, the second pitch lies between the first and third pitches, and the interval between the first and third notes is greater than MaxPrac or less than MinPrac; all other changes are half changes.*

If the interval between the first and third notes in a group of three is too wide or narrow to have been played comfortably without the aid of an intervening finger, a change of hand position may be said to have occurred. Example: The first note in a melodic fragment is played by Finger 2 and the third by Finger 3. A position change occurs if the interval between the first and third notes is greater than 3 semitones (MaxComf) or less than 1 semitone (MinComf). The model deems a *full change* to occur if (a) the interval between the first and third notes is greater than 5 semitones (MaxPrac) or less than 1 semitone (MinPrac = MinComf in this case), (b) the second note is played by the thumb, and (c) the second pitch lies between the other two pitches. The change would be regarded as a *half change* if the interval was 5 or 4 semitones, if the second note was not played by the thumb, or if the second pitch lay outside the two pitches. In this case, the last two conditions would be satisfied simultaneously if the second finger was 5 and the second pitch was higher than the other two pitches. The effect of the position-change-count rule within the algorithm as a whole is to favor fingerings in which the number of changes of hand position is minimized or fingerings in which relatively long sequences of notes are played within a single hand position.

5. *Position-change-size rule: If the interval spanned by the first and third notes in a group of three is less than MinComf, add the difference between the interval and MinComf (expressed in semitones) to the estimated difficulty. Conversely, if the interval is greater than MaxComf, add the difference between the interval and MaxComf to the estimated difficulty.*

To economize on unnecessary movements, changes of hand position tend to involve as little hand and finger motion as possible. Here, we take the distance in semitones traveled by the hand's center of gravity during a change of hand position as a measure of the difficulty of the change. Example: For F4–G4–A4 played 3–1–2, the position-change-size rule generates 5 points, because the fingers move about 5 semitones during the change, which comprises the interval between F4 and A4 (4 semitones) and MinComf for Fingers 3–2 (1 semitone).

6. *Weak-finger rule: Add one point to the estimated difficulty every time Finger 4 or Finger 5 is used.*

Fingers 4 and 5 are less strong and agile than Fingers 1, 2, and 3. Example: The repeating sequence B4–D5–E5–D5 is easier to play 1–2–3–2 (no points per cycle) than 2–4–5–4 (3 points per cycle).

7. *Three-four-five rule: Add one point to the estimated difficulty every time Fingers 3, 4, and 5 occur consecutively in any order, even when groups overlap.*

Runs of notes are difficult if played by fingers on the weak side of the hand. Example: For the first 8 notes of Piece A played 3–5–4–5–3–4–2–3, the three-four-five rule assigns 3 points, because embedded within this fingering are the consecutive sequences 3–5–4, 4–5–3, and 5–3–4. These points are in addition to points already generated by the weak-finger rule.

8. *Three-to-four rule: Add one point to the estimated difficulty each time Finger 3 is immediately followed by Finger 4.*

9. *Four-on-black rule: Add one point to the estimated difficulty if Fingers 3 and 4 occur consecutively in any order with Finger 3 on white and Finger 4 on black.*

The transition from Finger 3 to Finger 4 is difficult because it involves raising the fourth finger while the third is held down. In the present study, pianists used the fingering 4–3 significantly more often than 3–4 on consecutive notes of a single right-hand line.[A1] The three-to-four rule accounts for the difficulty involved in playing Finger 4 after Finger 3 regardless of whether the notes involved are black or white. Additional difficulties due to key elevation are accounted for by the four-on-black rule.

10. *Thumb-on-black rule: Add 1 point to the estimated difficulty whenever the thumb plays a black key. If the immediately preceding note is white, add a further 2 points. If the immediately following note is white, add a further 2 points.*

11. *Five-on-black rule: If the fifth finger plays a black key and the immediately preceding note is white, add 2 points. If the immediately following key is white, add 2 further points.*

Pianists tend to avoid using the thumb and (to a lesser extent) the fifth finger on black keys, because Fingers 1 and 5 are the shortest

[A1] Among some 30,000 fingered notes, the transition 4–3 occurred 1,663 times, whereas the reverse sequence 3–4 occurred 1,142 times. Assuming a binomial distribution, these two values are significantly different ($p < .01$). Assuming that rising and falling intervals of the same size occurred equally often in this database, this difference is consistent with the three-to-four rule. Other significant asymmetrical finger transitions observed in our data (3–1 > 1–3, 1–4 > 4–1, 3–2 > 2–3, 2–4 > 4–2, 5–4 > 4–5) were less pronounced (when measured as percentages) and are not accounted for by the model.

(*Appendixes continued*)

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of the five fingers, and the black keys are shorter than the white keys. The difficulty involved in placing short fingers on black keys depends on the context in which a note appears, because context can change the position of the hand relative to the keyboard. It is easier to play a black key with Finger 1 or 5 if the hand is already relatively distal ("into the keys"), which is the case if the preceding or following key is also black. Example: For D♭4–B♭4 played 1–5, the thumb-on-black rule generates 1 point for Finger 1 on D♭, but the five-on-black rule does not penalize Finger 5 on B♭, because both keys are black. For D4–B♭4 played 1–5, there is no penalty for Finger 1 on D, but 2 points are assigned for Finger 5 on B♭.

12. *Thumb-passing rule: Add 1 point to the estimated difficulty for each thumb pass on the same level (from white to white or from black to black). Add 3 points if the lower note is white and played by a finger other than the thumb and if the upper note is black and played by the thumb.*

The thumb-passing rule accounts for the role of key elevation in thumb passing. The thumb-passing and thumb-on-black rules combine to account for the difficulty of thumb passes that involve placing the thumb on a black key. Example: For A#3–C#4–E4 played 2–1–2, the thumb-on-black rule assigns 3 points, and the thumb-passing rule assigns 1 point, for passing from nonthumb on black to thumb on black (i.e., passing on the same level). For A3–C#4–E4 played 2–1–2, the thumb-on-black rule again generates 3 points, and the thumb-passing rule assigns 3 points, for passing from nonthumb on white to thumb on black.

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Appendix B

The Seven Pieces Used in the Experiment

The following seven studies by C. Czerny used in this experiment are from *160 Kurze Übungen* [160 Eight-Bar Exercises] *Opus 821,* 1845?/1920, London: Peters Edition Limited. Copyright 1920 by C. F. Peters; copyright renewed 1948 by C. F. Peters. Reprinted with permission.

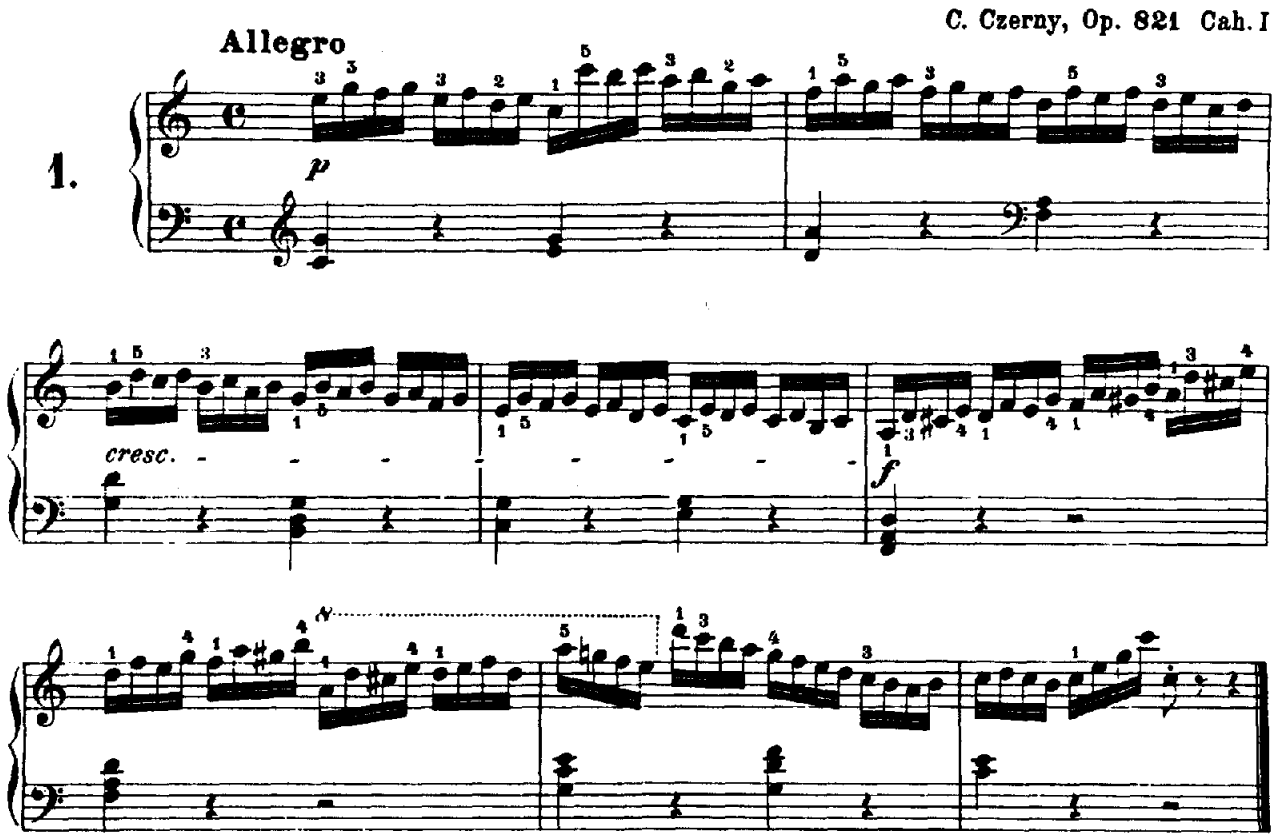

Figure B1. No. 1 in C Major (Piece A).

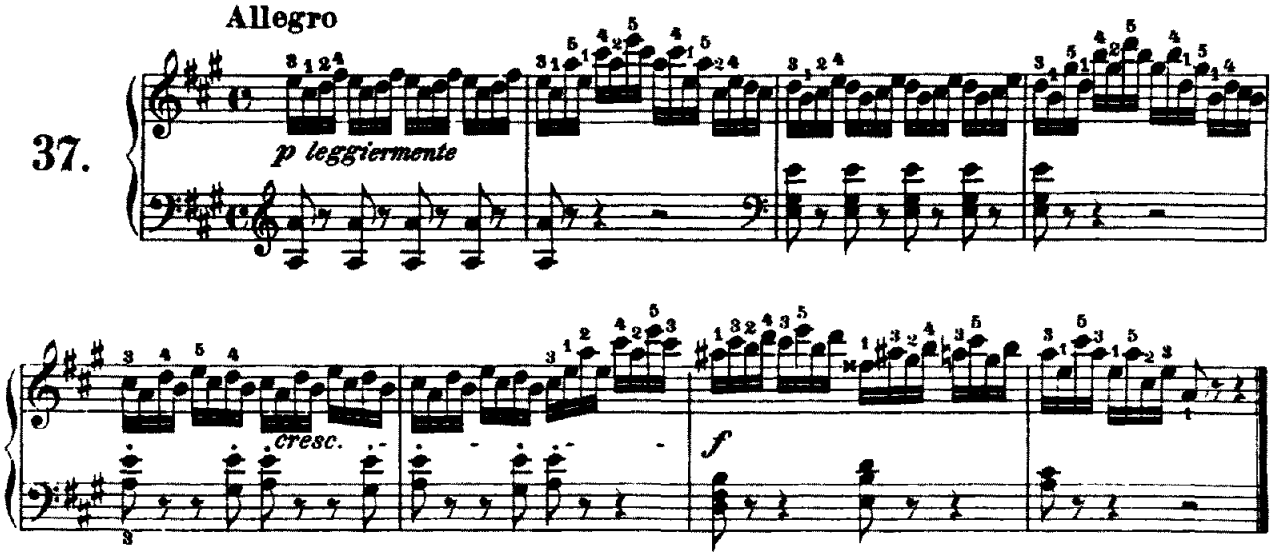

Figure B2. No. 37 in A Major (Piece B).

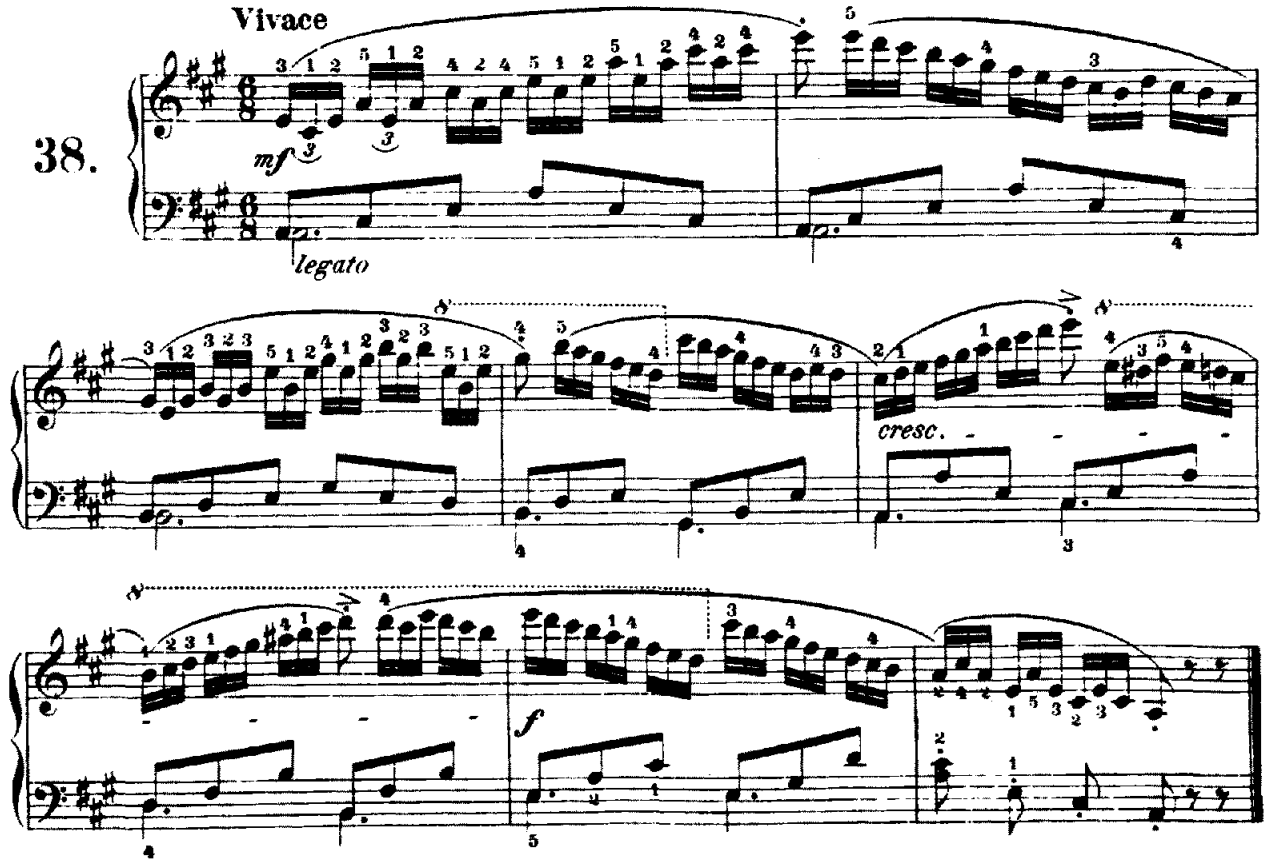

Figure B3. No. 38 in A Major (Piece C).

(*Appendixes continued*)

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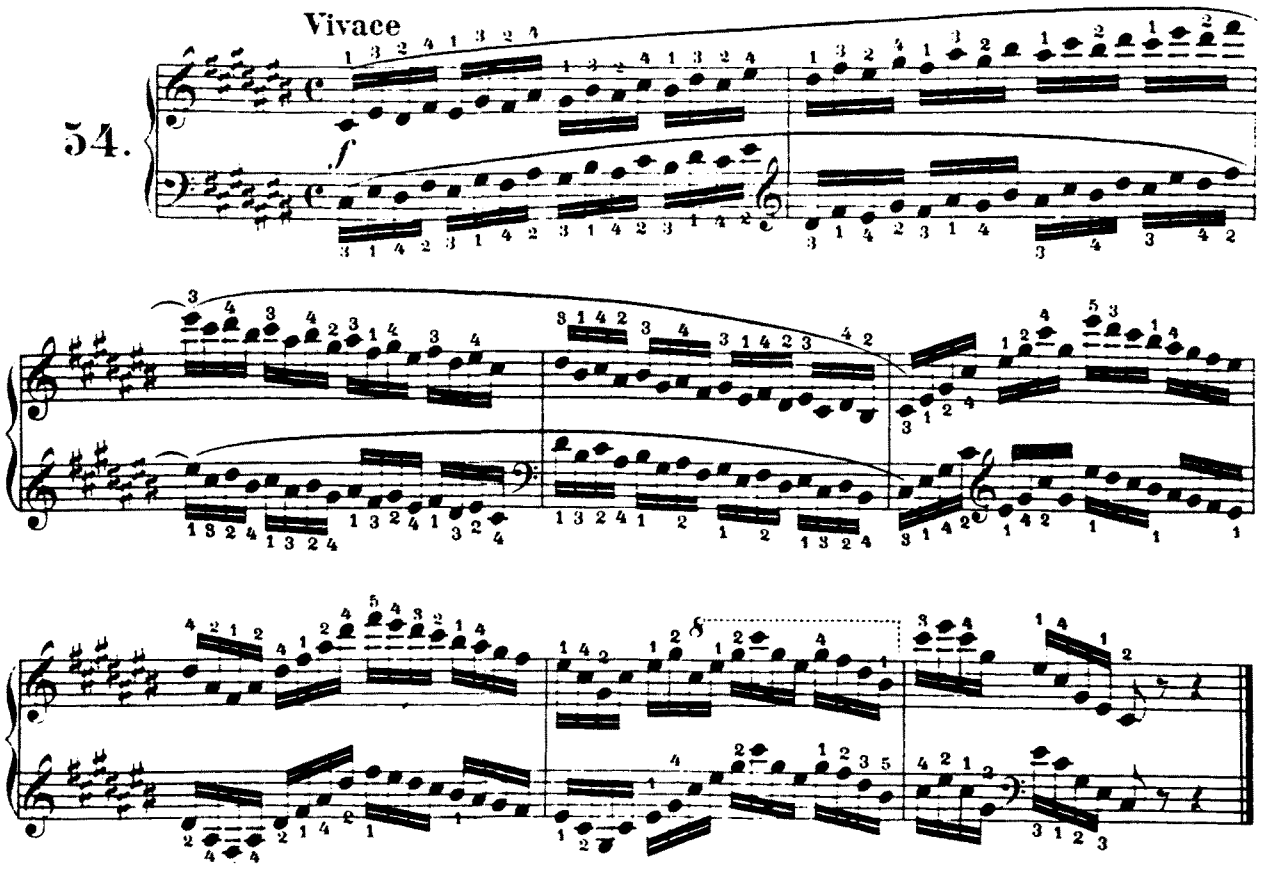

Figure B4. No. 54 in C# Major (Piece D).

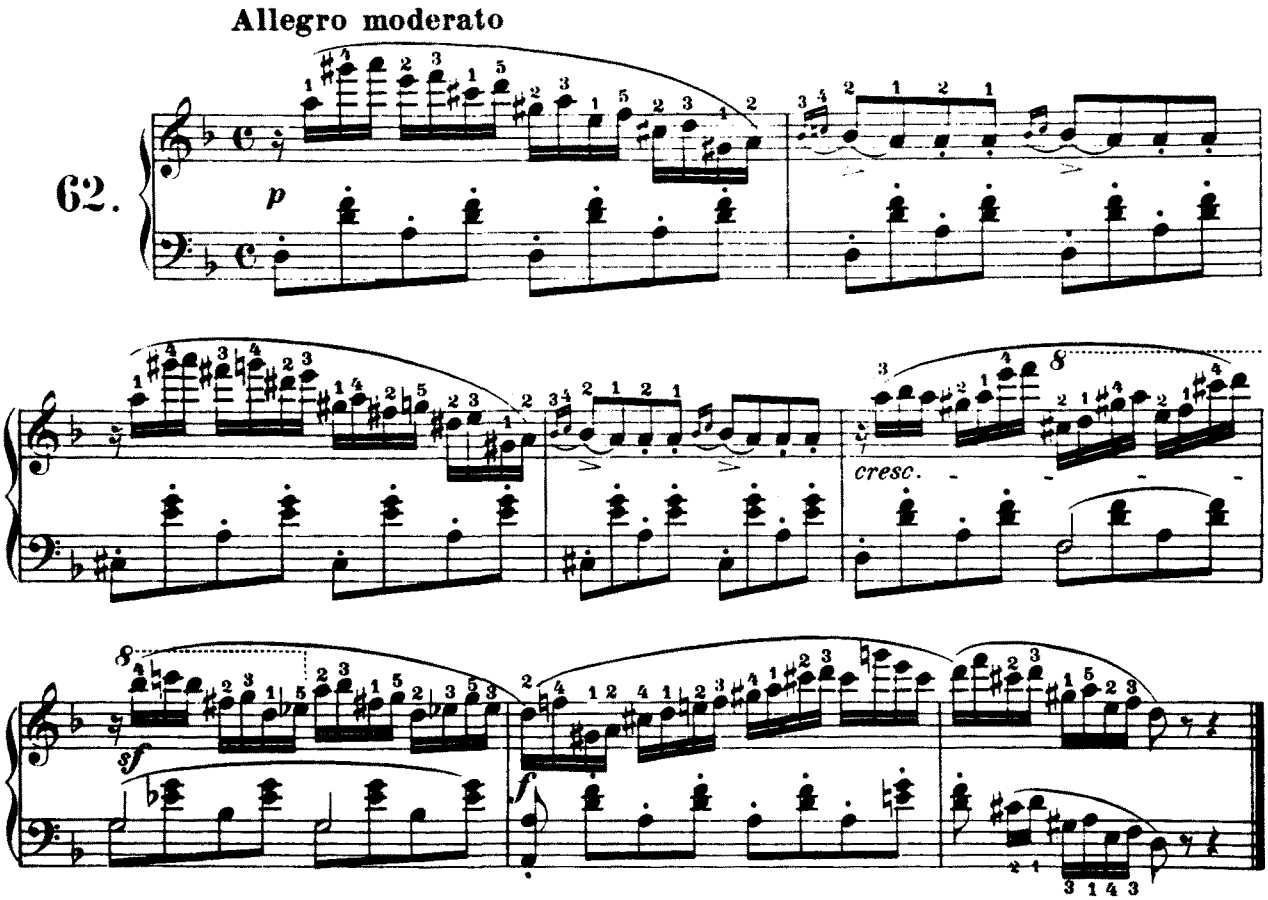

Figure B5. No. 62 in D Minor (Piece E).

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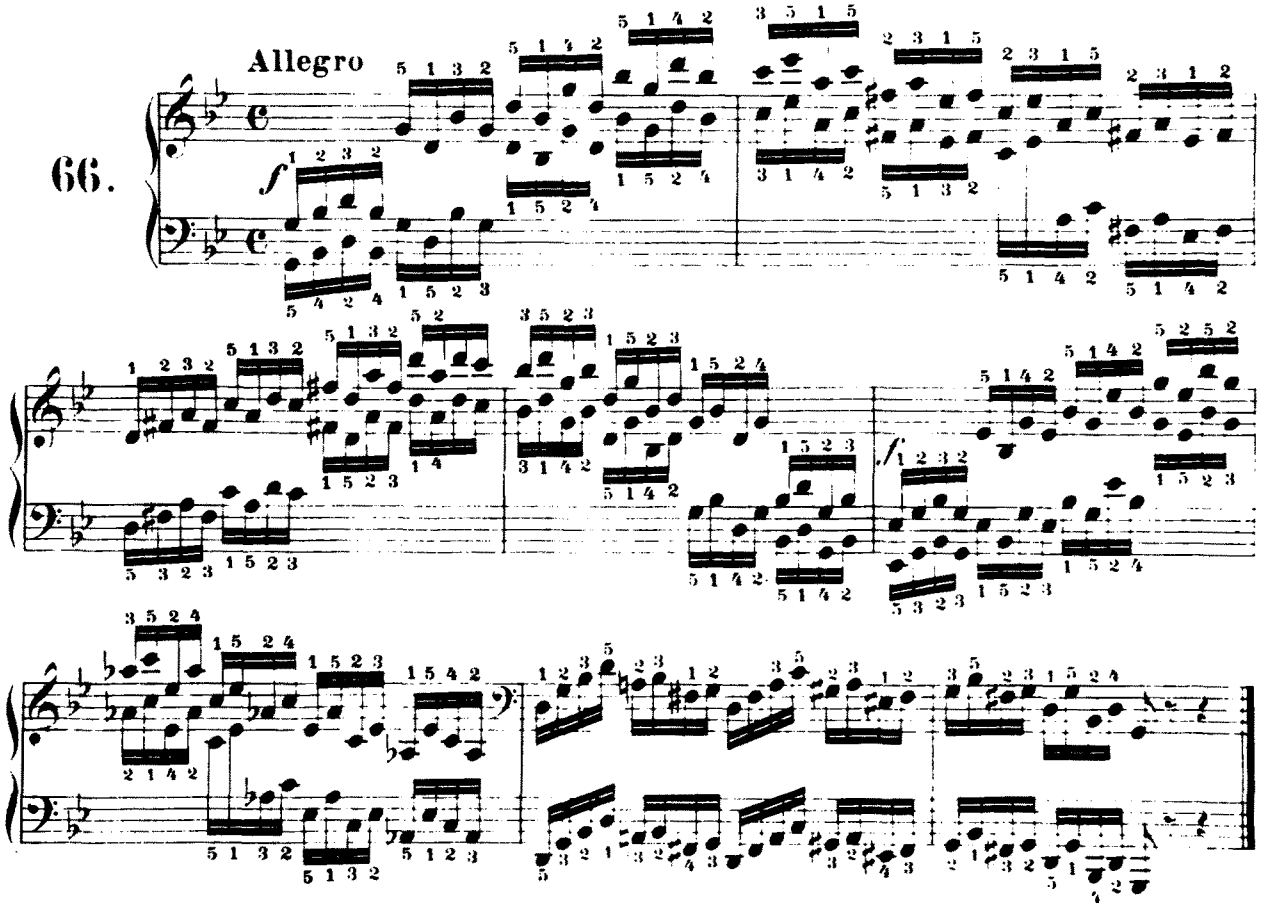

Figure B6. No. 66 in G Minor (Piece F).

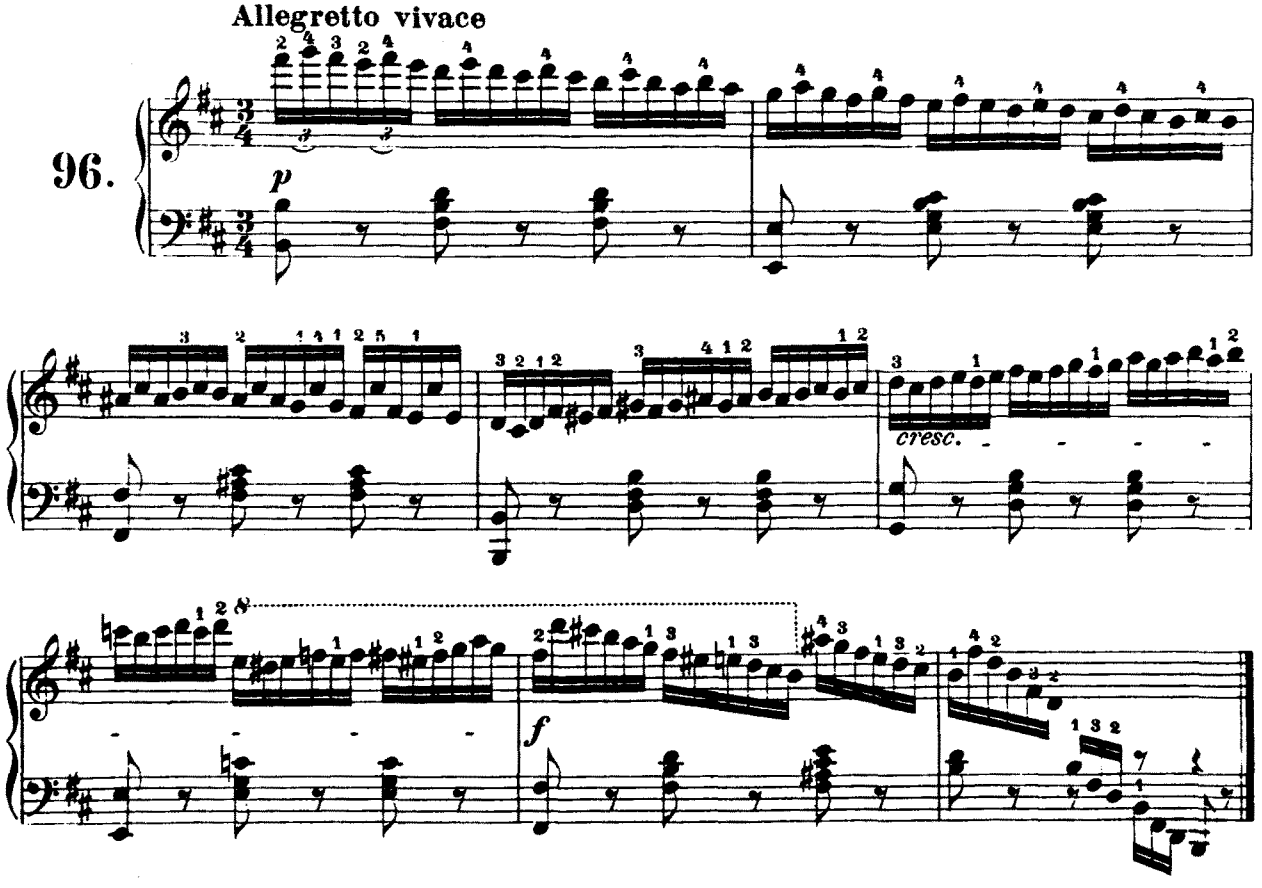

Figure B7. No. 96 in B Minor (Piece G).

(*Appendixes continued*)

Appendix C

Algorithms for Error Detection

The process of eliminating note errors is not as straightforward as it may first seem. If only one note is added or removed from one of the performances and it is embedded in a sequence of notes that was otherwise performed accurately in both cases, the location of the error can generally be determined unambiguously. But pianists often omit or add whole sequences of successive notes. This, combined with the fact that the pieces under consideration often include short repetitive patterns—sequences whose cycle length can approach the length of note sequences omitted or added by pianists in performance—means that the exact location of errors is often ambiguous, and more than one feasible solution may exist that will render the two passages identical.

Given the inherent ambiguity of the note-elimination process, we developed two contrasting algorithms for the detection of errors and compared results obtained with each. We call them the recursive and nonrecursive algorithms.

Before applying either algorithm, we first checked the actual performances by listening to them. Where a pianist had repeated a relatively large section of music (say, 6 or more notes), we eliminated the first repetition and retained only the second.

The recursive algorithm works in the following way. The two takes of the same piece are searched from the beginning until a pitch mismatch is found. A number of possible deletions are then tried out. They involve deleting between 1 and 6 notes, either from the first or from the second take (the number 6 was chosen arbitrarily to correspond roughly to the hand–eye span). For each possible deletion, the agreement is calculated between the pitches of the two takes within a 6-note window (beginning at the position of the original mismatch). Whichever deletion produces the best agreement is then adopted. If two or more solutions give the same level of agreement, then the solution with the minimum number of deletions is adopted. If none of the $6 \times 2 = 12$ possibilities produces better agreement than the original version (i.e., the removal of no notes at all), then one note is deleted from each take at the position of the mismatch (i.e., at the start of the window). Once a decision has been reached regarding which notes to delete, they are actually deleted, and the program returns to the same position as before to check the identity of the note in each take. The procedure continues until the end of the piece is reached. This procedure is recursive in the sense that it checks the consequences of each deletion before actually performing it.

The nonrecursive algorithm takes longer to explain, but owing to the absence of recursions it is more straightforward to program and runs more quickly. As before, each note in the first take is compared with each note in the second take until a pair is found for which the pitches are not the same. A window is then selected from each take, beginning with the noncoinciding pitches and extending for 10 notes (the results of the procedure depend relatively little on the size of this number; we considered that a nonrecursive procedure should look a little further ahead than a recursive procedure). A 10×2 matrix of "signed minimum time intervals" is then calculated. These indicate, for each of the 10 chosen notes in each take, the distance (expressed as the number of intervening notes) between that note and the nearest note in the other take with the same pitch. When looking for other notes with the same pitch, we confined the search to a window beginning with the original noncoinciding pitches and extending for 20 notes. A plus sign or a minus sign is assigned to each value in the matrix to indicate whether the nearest note in the other take lies ahead or behind the note in question, respectively. If no other note is found with the same pitch, an arbitrarily large value is assigned (here, 99).

Each time a note error is found, three possibilities are considered for correcting it: the deletion of one note from the first take, the deletion of one note from the second take, or the deletion of one note from each take. The decision is based on the following criteria.

1. If all entries in the matrix of signed minimum time intervals are zero except for the first column, both of the noncoinciding pitches are deleted. This case applies to simple note substitution errors. If, for example, a scale is first played correctly and then played with a mistake on the first note,

C4 D4 E4 F4 G4 A4

E4 D4 E4 F4 G4 A4,

then the matrix will be

99 0 0 0 0 0

+2 0 0 0 0 0.

The nonzero entries show clearly which notes are to be deleted to make the two series identical (here, the C4 and the first E4). (Note that without consulting the score, we do not know in which take the "mistake" really occurred.)

2. The second case applies to a simple insertion or deletion (note again that without consulting the score there is no way of distinguishing these). If one of the numbers in the first column of the matrix is the "big" number (here, 99), then only one note (the note corresponding to that number) is deleted. For example,

C4 D4 E4 F4 G4 A4

B3 C4 D4 E4 F4 G4

results in the matrix

+1 +1 +1 +1 +1 +1

99 −1 −1 −1 −1 −1.

Here, the first few notes of the two windows may be rendered identical by eliminating the B3.

3. If neither of the above procedures can be applied, then a note is deleted from the take with the largest number of negative entries in the window. As shown in the previous example, negative values suggest that agreement is more likely if the corresponding notes are shifted to the left.

4. If the number of negative entries is the same in both windows, a note is deleted from the take in which the negative entries add to the largest negative sum. Again, this deletion is supposed to be most likely to lead to an eventual agreement between the two takes.

5. If all else fails, one note is deleted from that take from which a note was not deleted last time. This prevents too many notes from being deleted from one take, given that the two takes were originally of about the same length.

As soon as one of the above criteria is found to apply, a decision

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is immediately reached regarding the note to be deleted, and later criteria in the list are ignored. Most note deletions are determined by the first two criteria; the last three are required less often. Once a note has been deleted, the program returns to the same note position and again checks the identity of the two pitches. If they are still different, the matrix is recalculated and the same procedure is applied again.

We have found the nonrecursive routine to work well for pianists who make only one error at a time and do not go back and correct themselves. However, it tends to delete more notes than the recursive routine when pianists omit or repeat successive notes. Neither routine can handle note reversals; if two notes are played in two different orders, then one of the two will be deleted in each take.

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